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(54) **HYBRID ENERGY SYSTEMS**

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(71) Applicants: **Alliance North America, Inc.**, Cypress, CA (US); **SPOC Grid Inverter Technologies, Inc.**, Trussville, AL (US)

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(72) Inventors: **Michael Yin**, Santa Ana, CA (US);
Brian Quock, Huntington Beach, CA (US); **Stacey Brooks**, Boaz, AL (US);
Christian Mason, Trussville, AL (US);
Ben Gully, Anacortes, WA (US);
Andrew Williams, Pass Christian, MS (US)

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(57) **ABSTRACT**

A hybrid energy system is configured to carry a power load for a generator configured to output a first AC signal. The hybrid energy system includes a battery bank, a DC/DC converter, an AC/DC converter, and a DC/AC converter. The battery bank includes a plurality of batteries and outputs a first DC signal. The DC/DC converter, operating in a first mode, receives and converts the first DC signal into a second DC signal, which is output to a DC bus. The AC/DC converter receives and converts the first AC signal into a third DC signal. The second DC signal and the third DC signal are tied together on the DC bus. The DC/AC converter receives and converts the second DC signal from the DC bus into a plurality of second AC signals, which are output to an AC outlet interface.

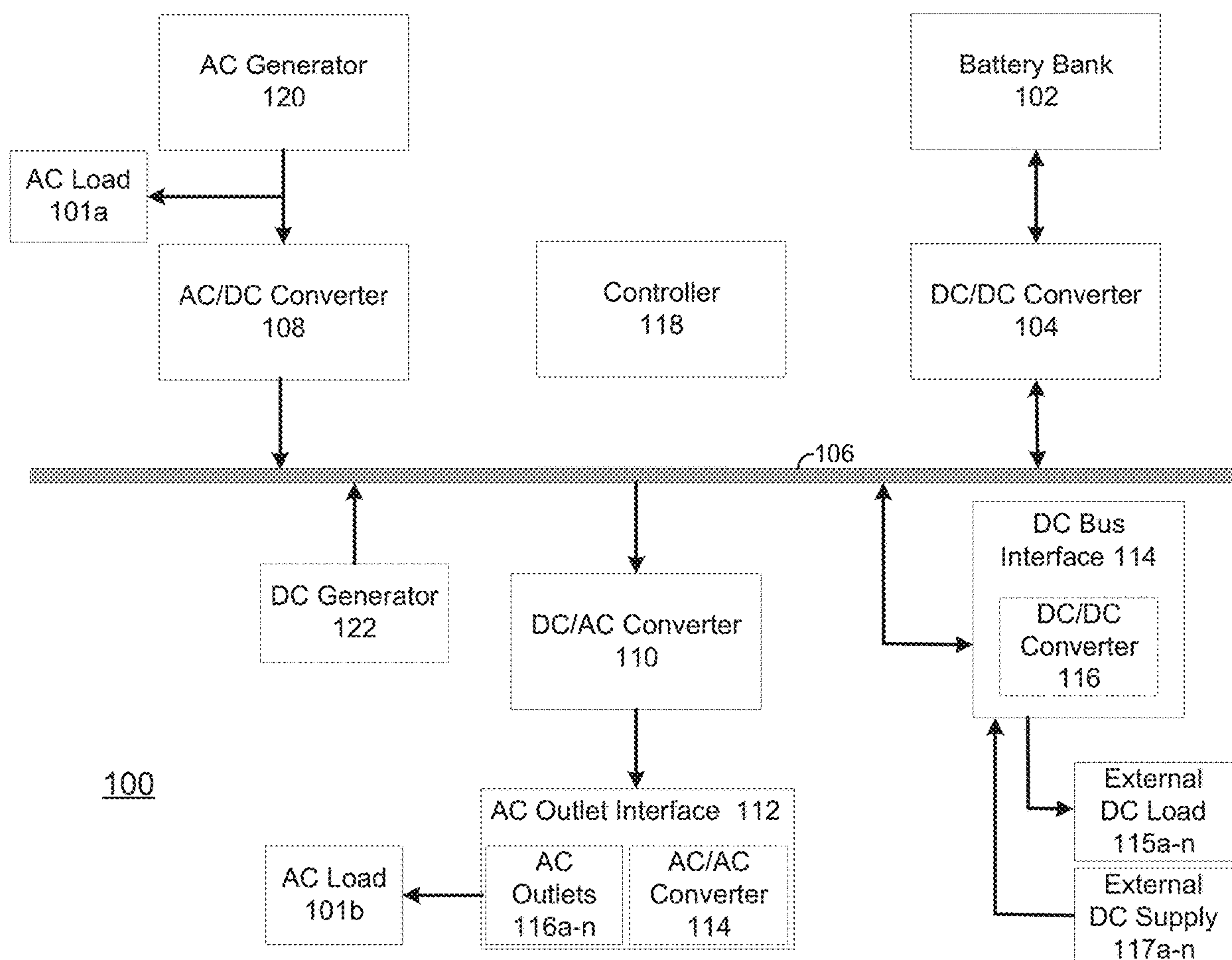
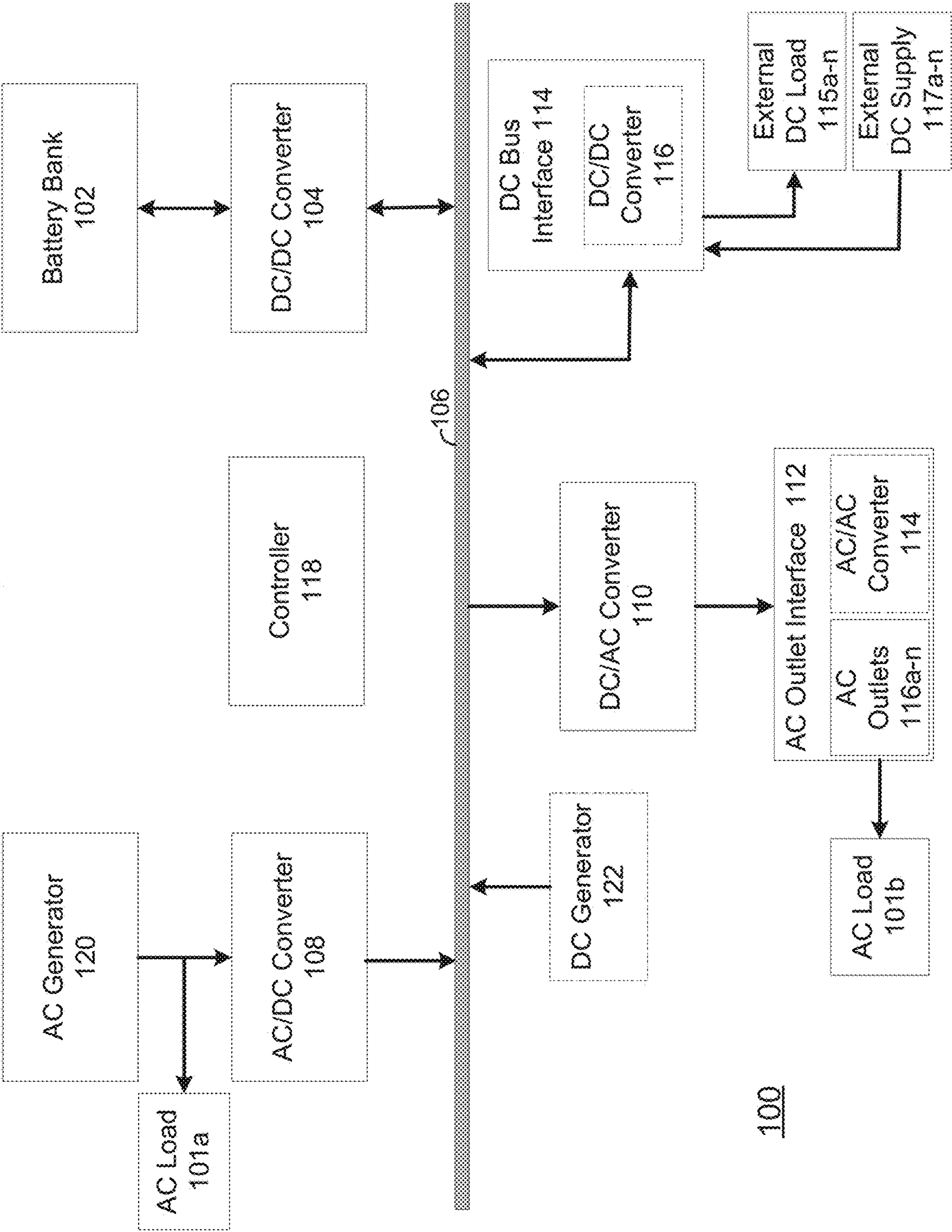
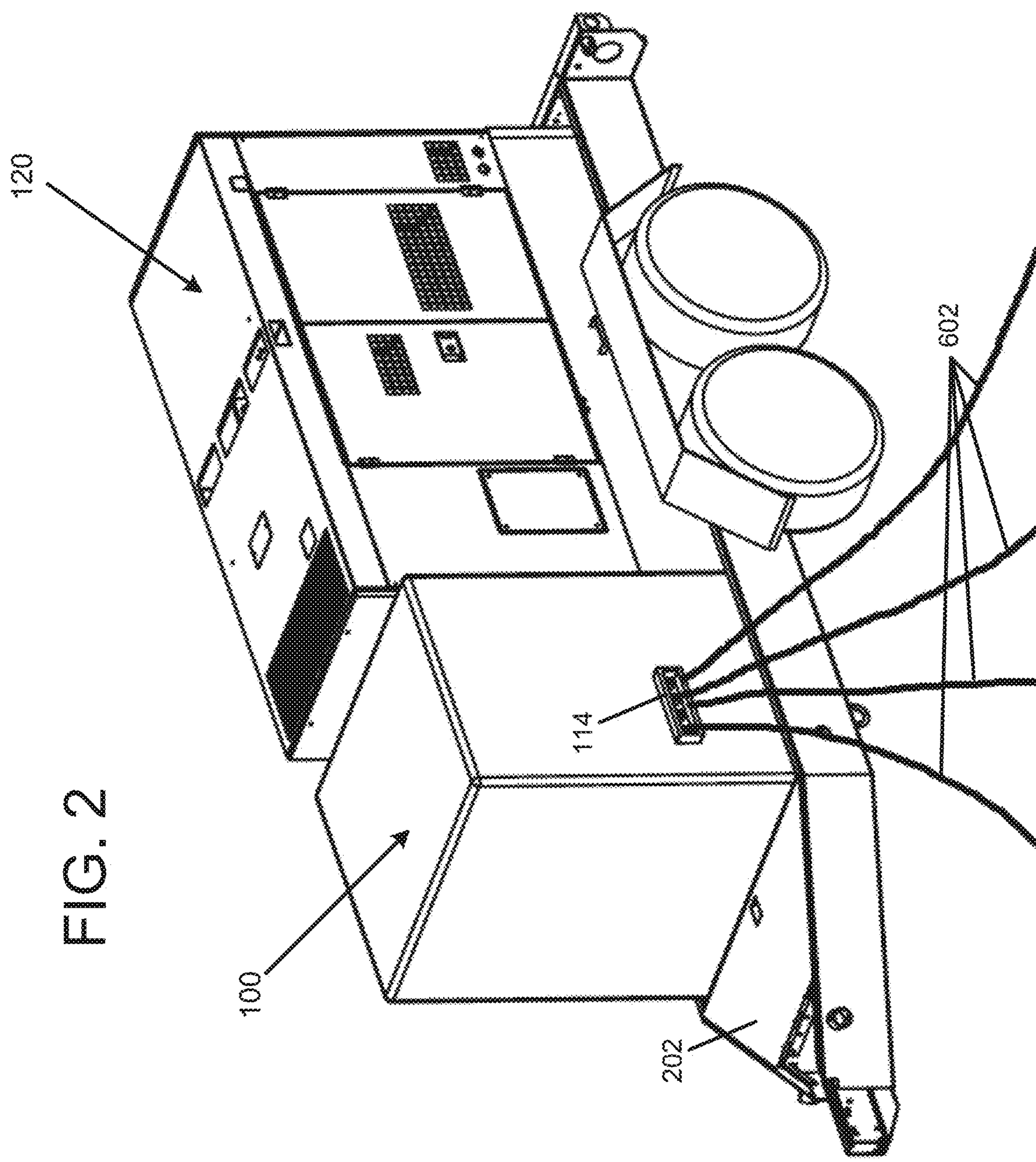
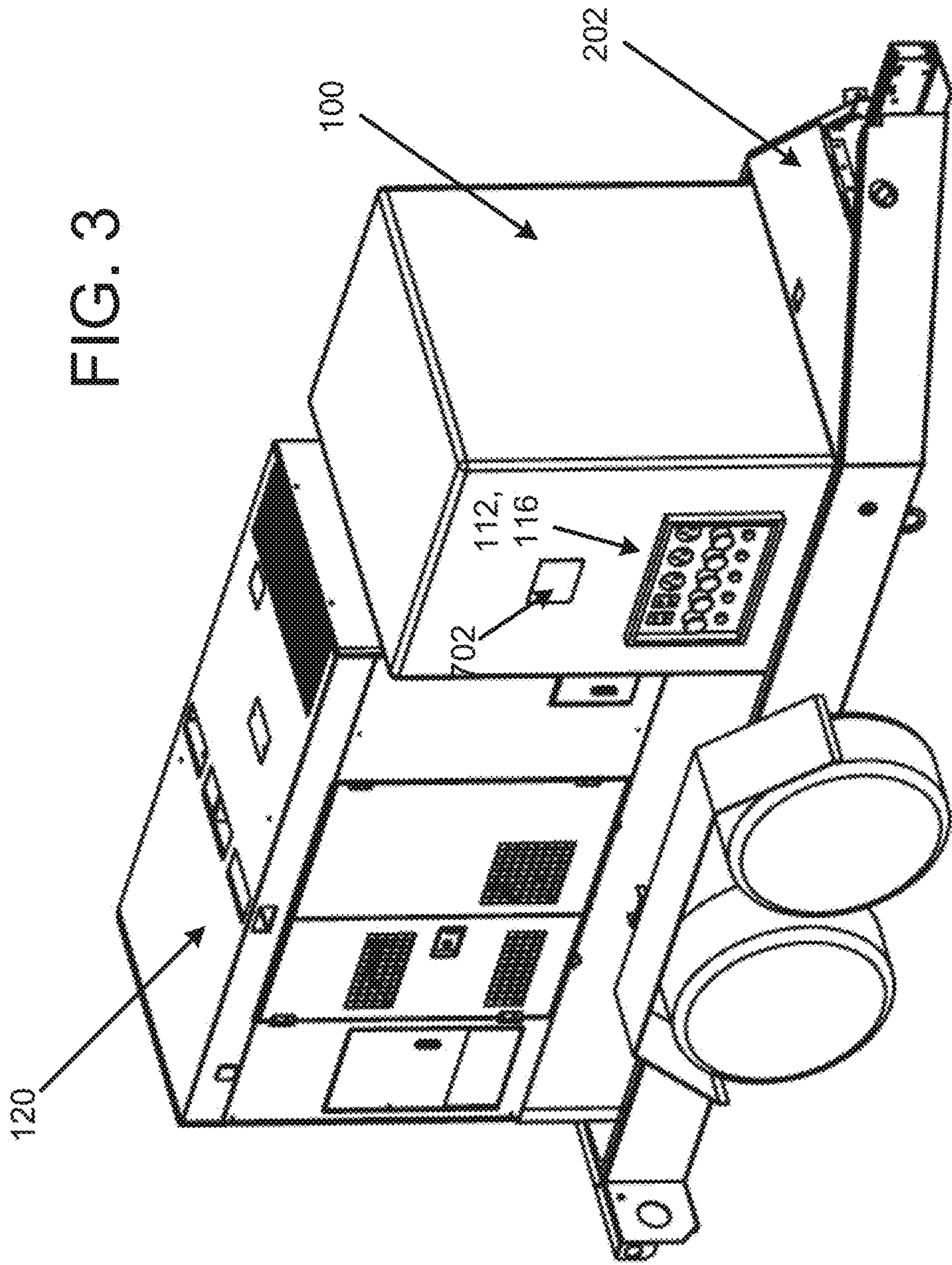


FIG. 1







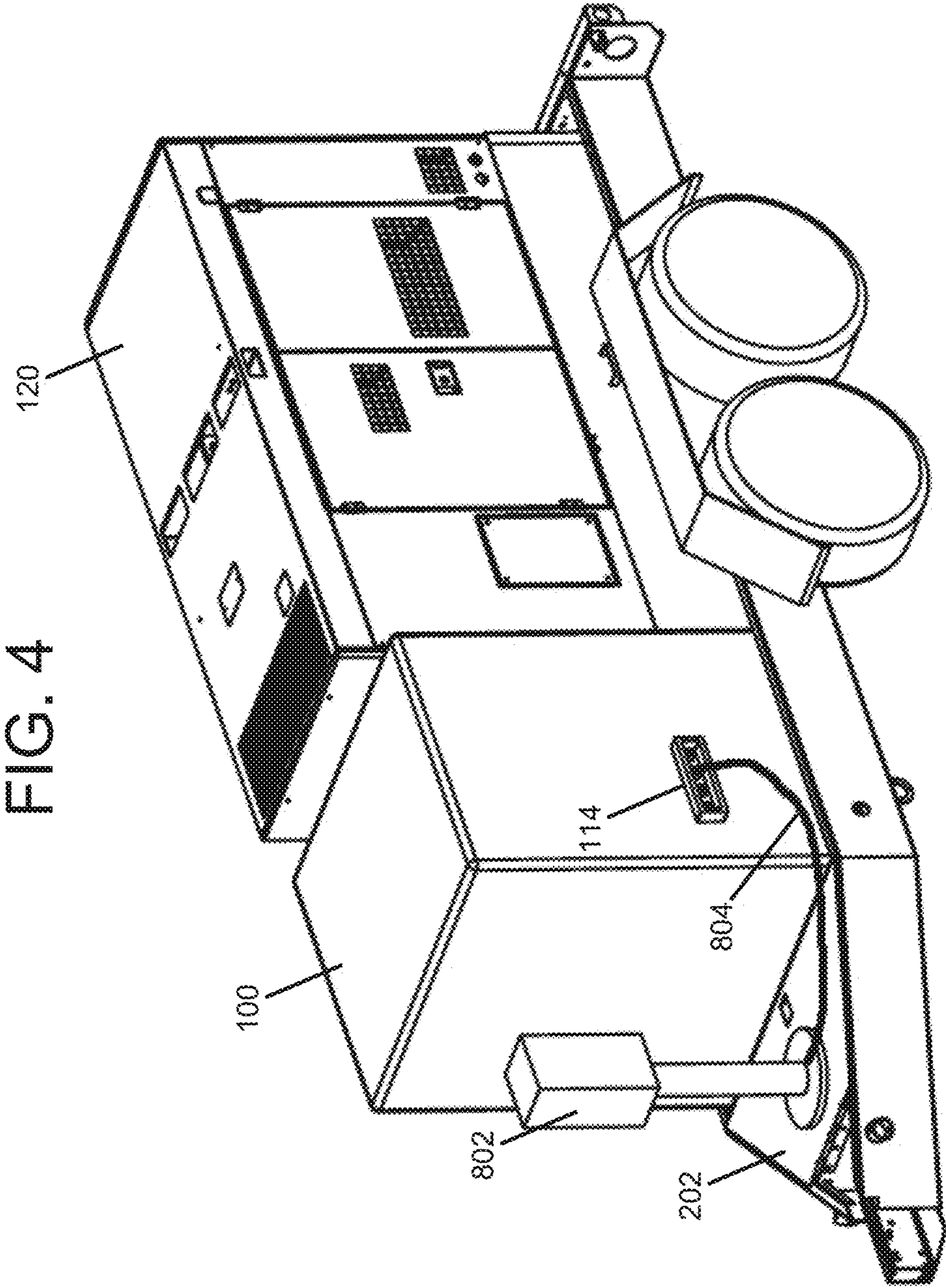


FIG. 5

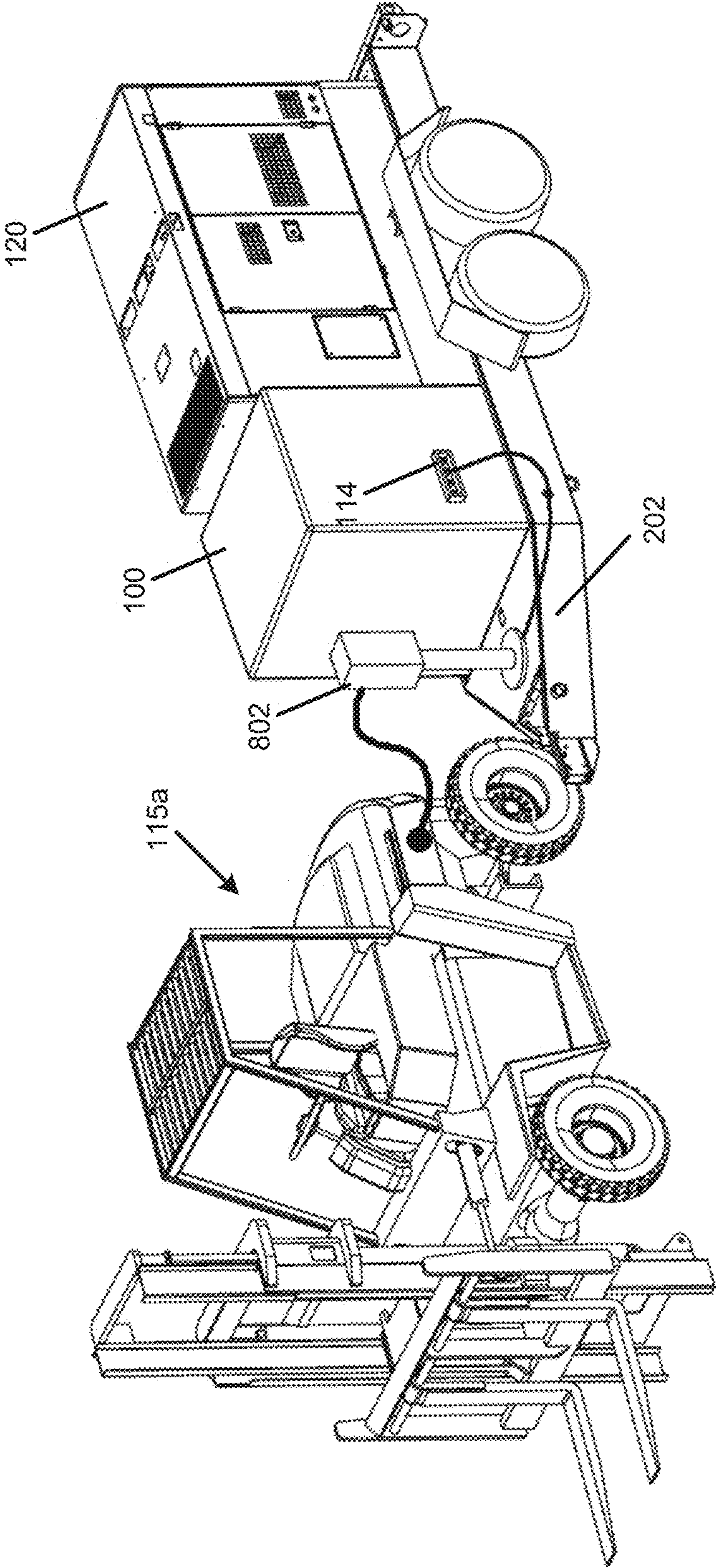


FIG. 6A

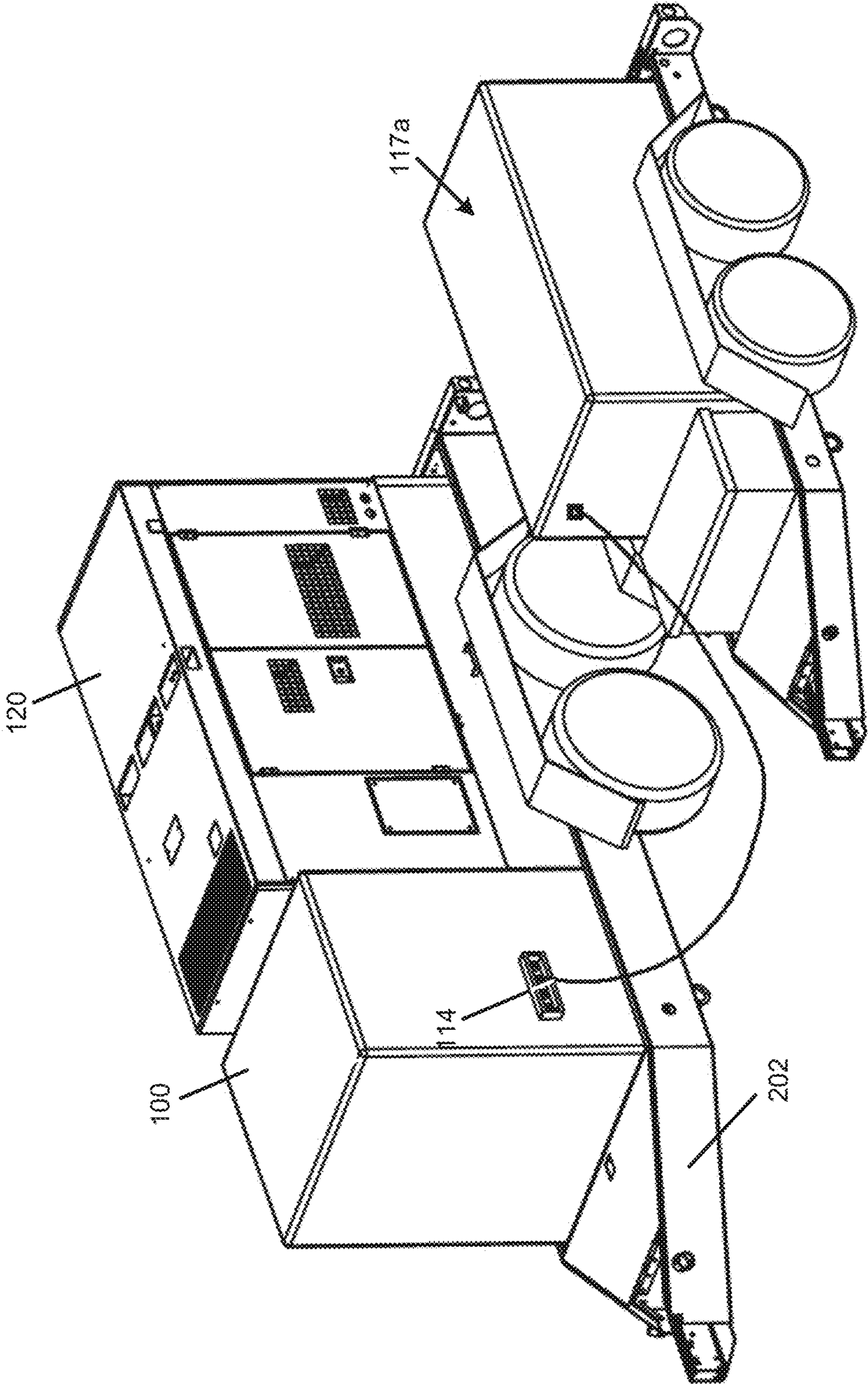


FIG. 6B

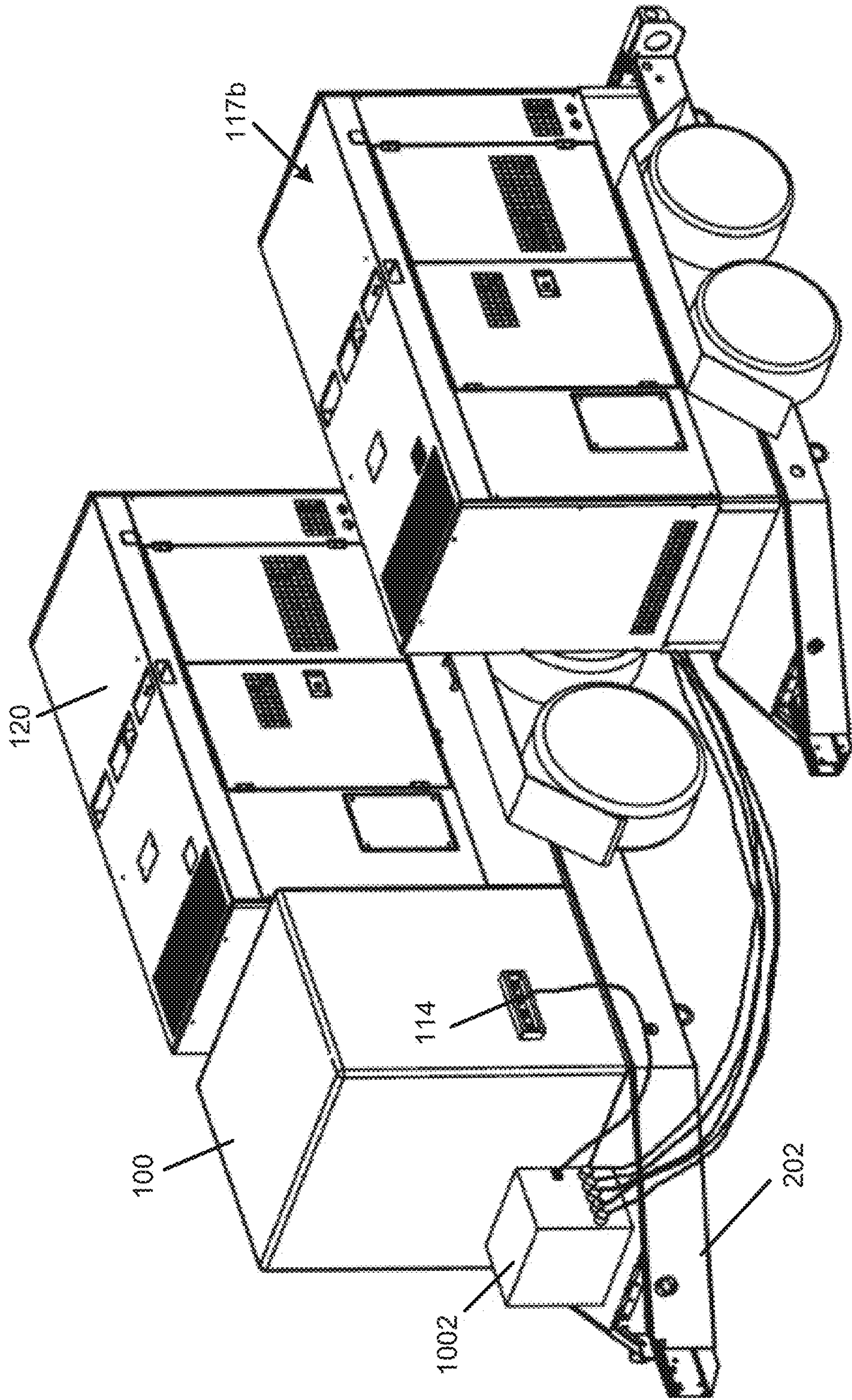


FIG. 6C

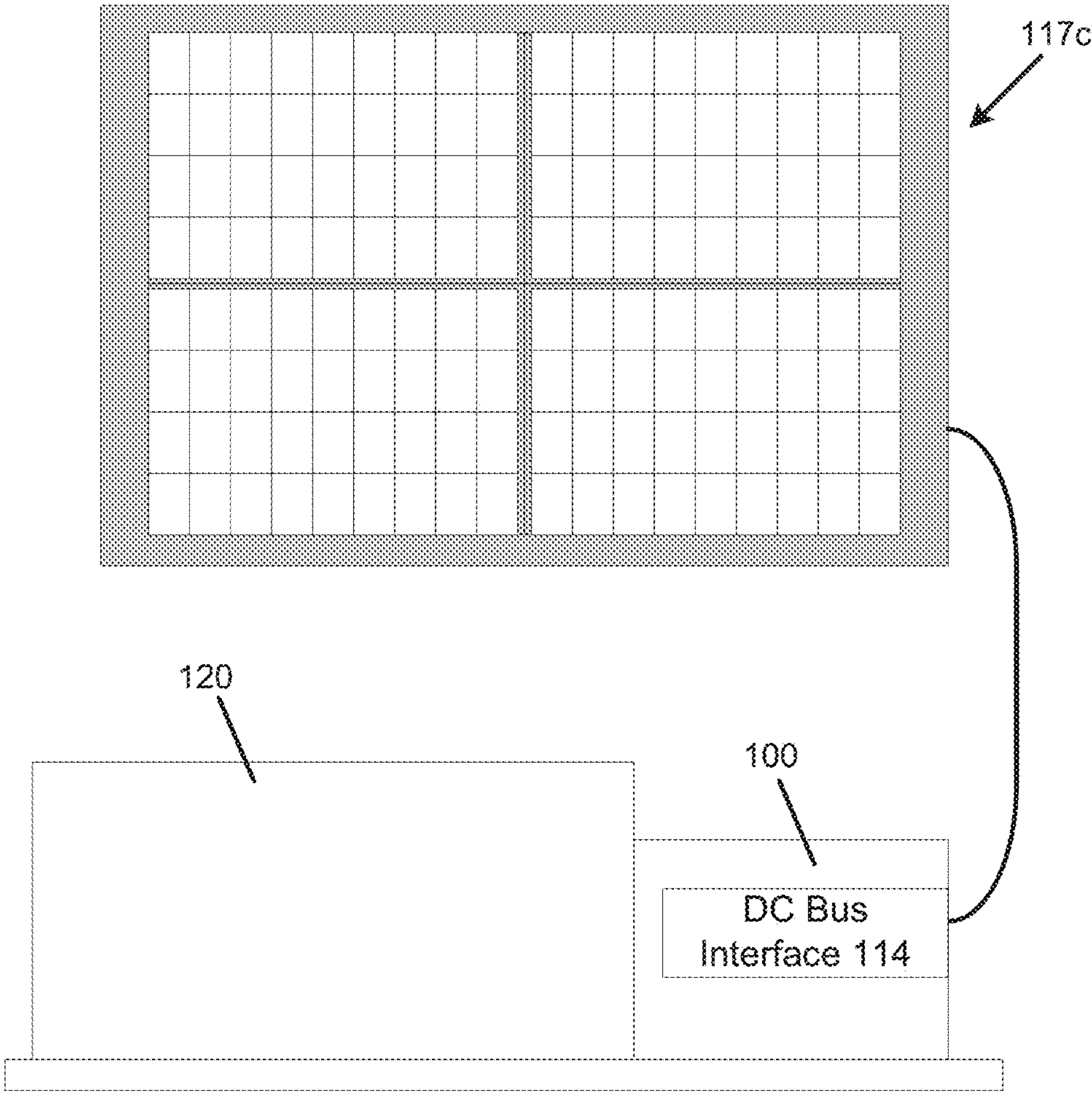


FIG. 6D

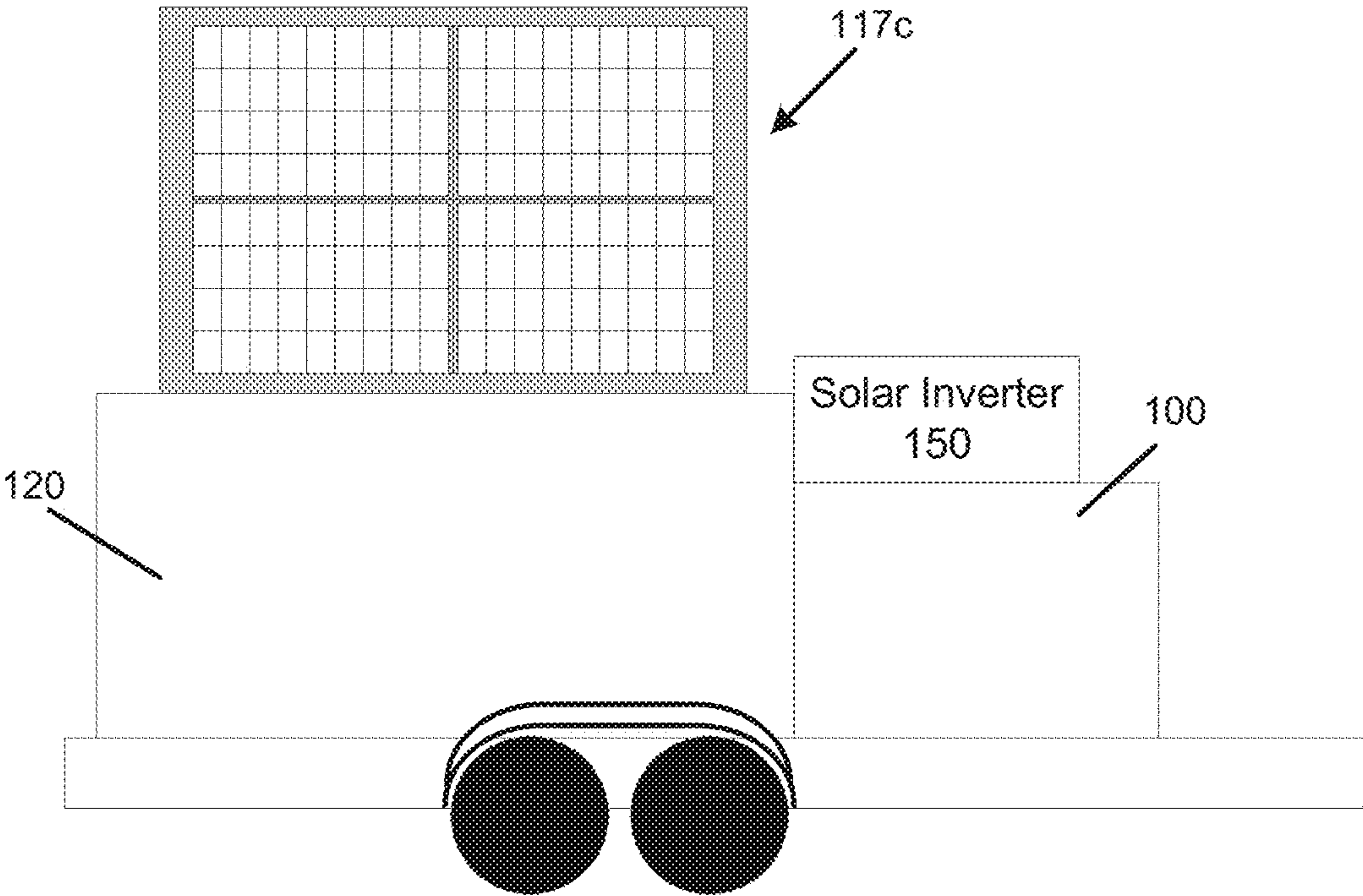


FIG. 6E

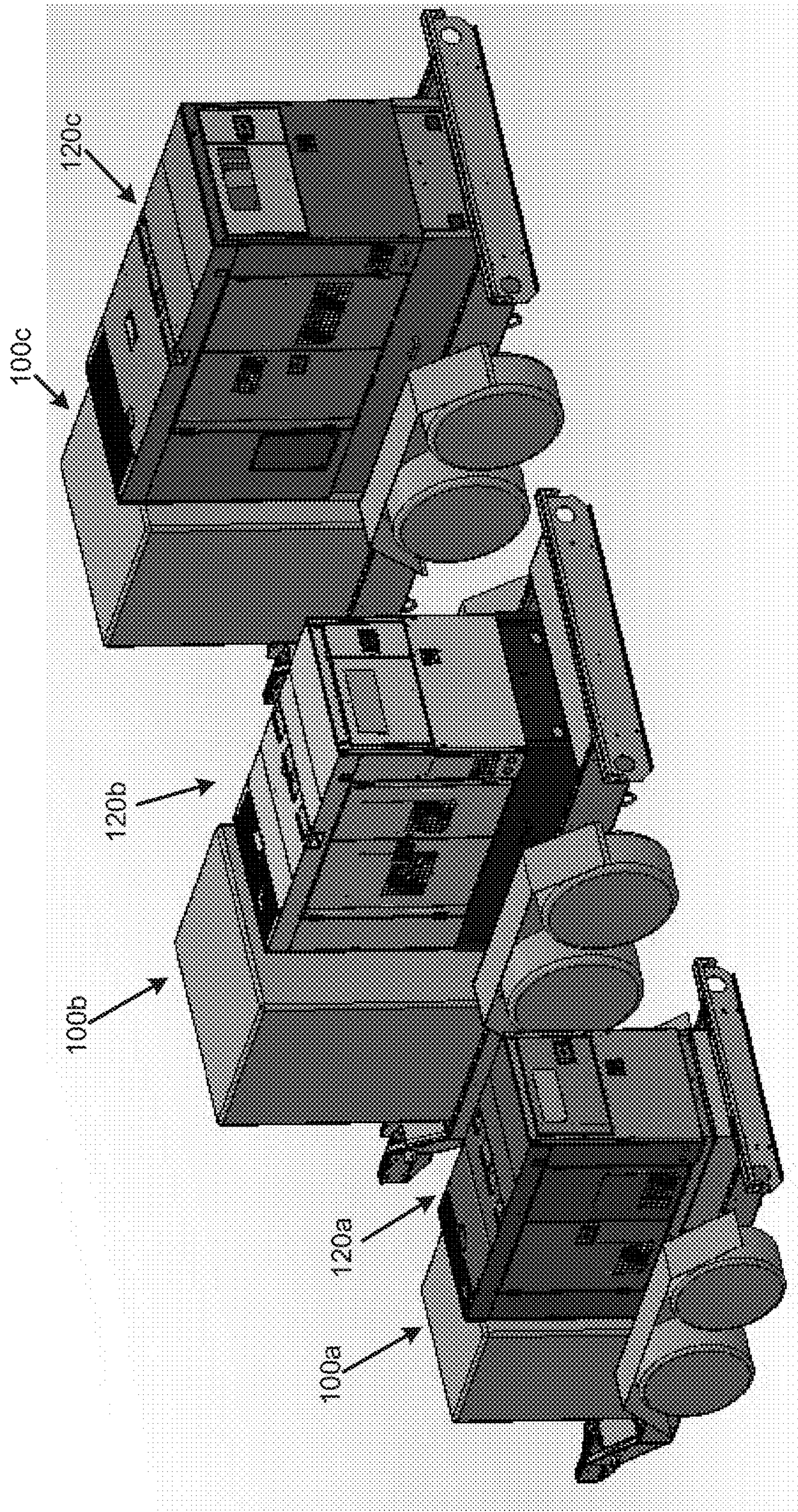


FIG. 6F

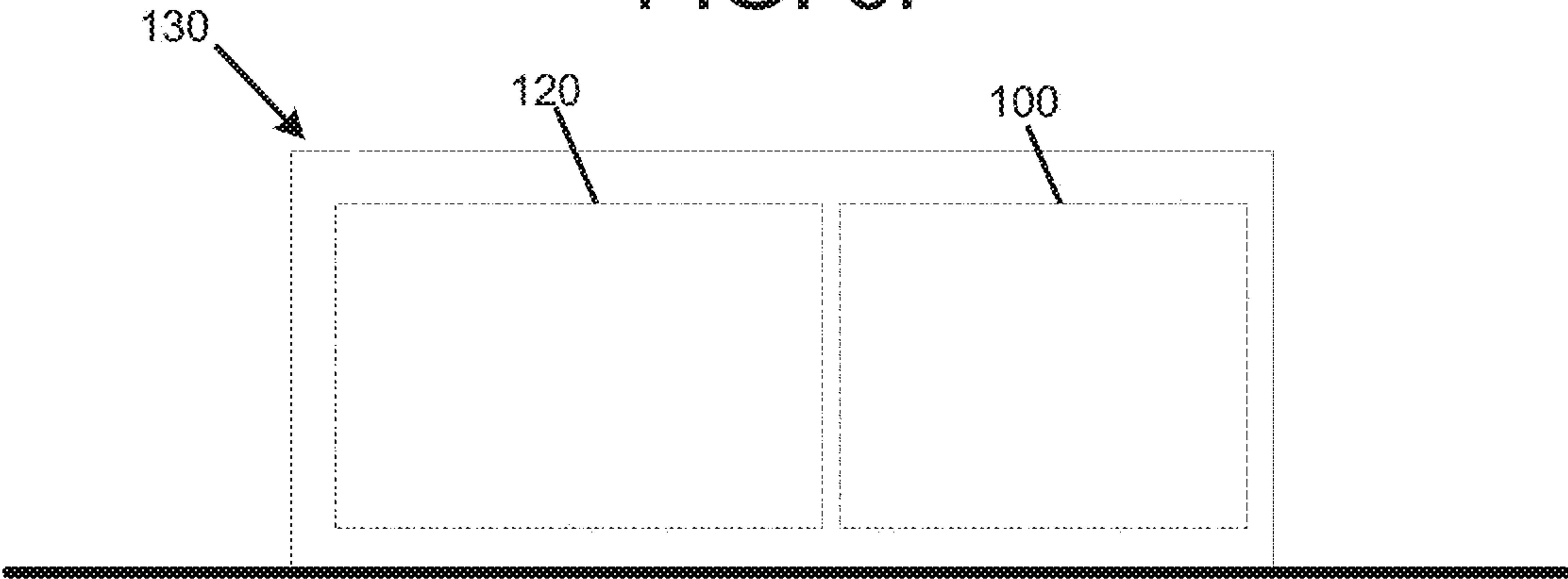


FIG. 6G

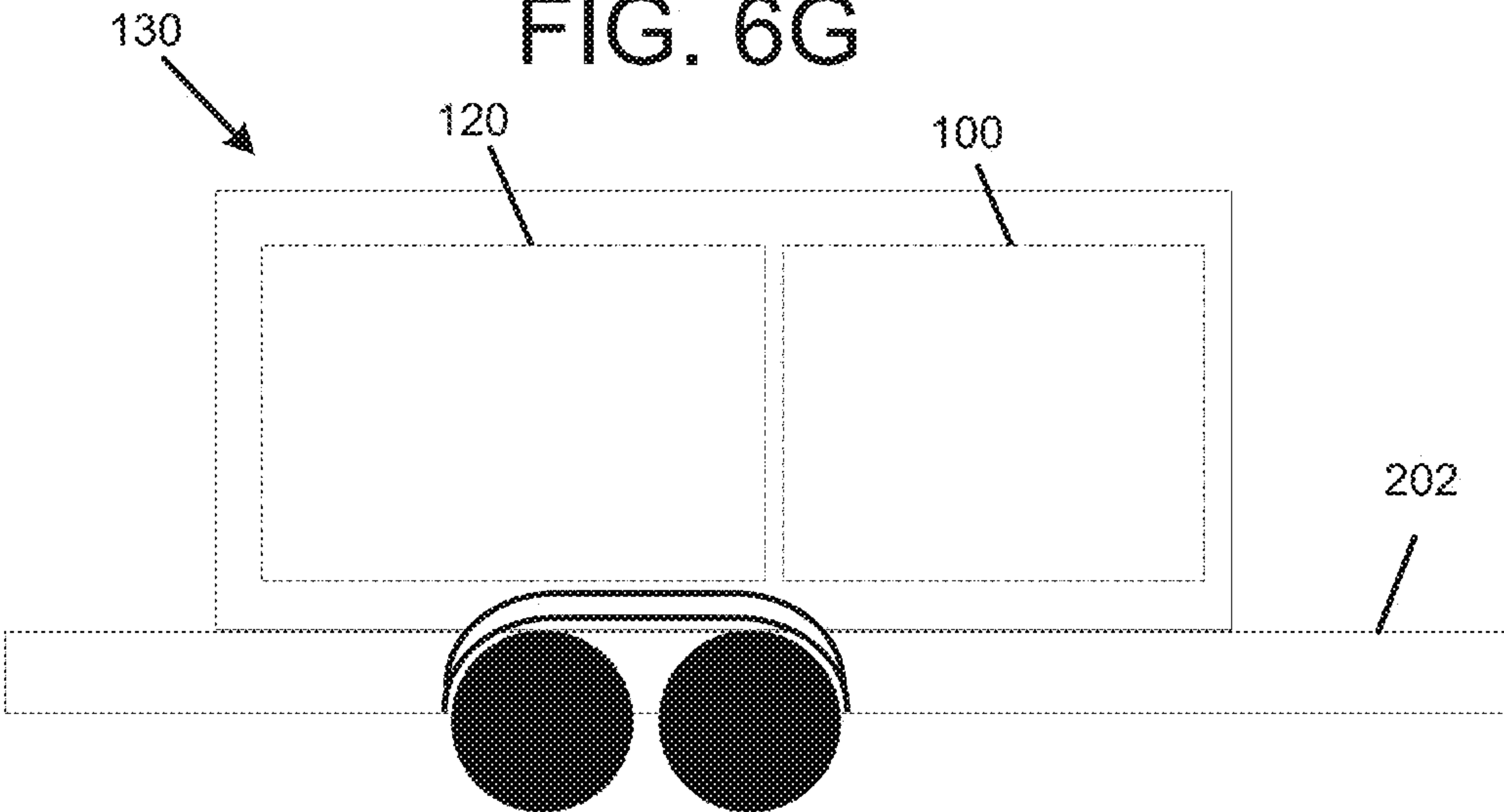


FIG. 6H

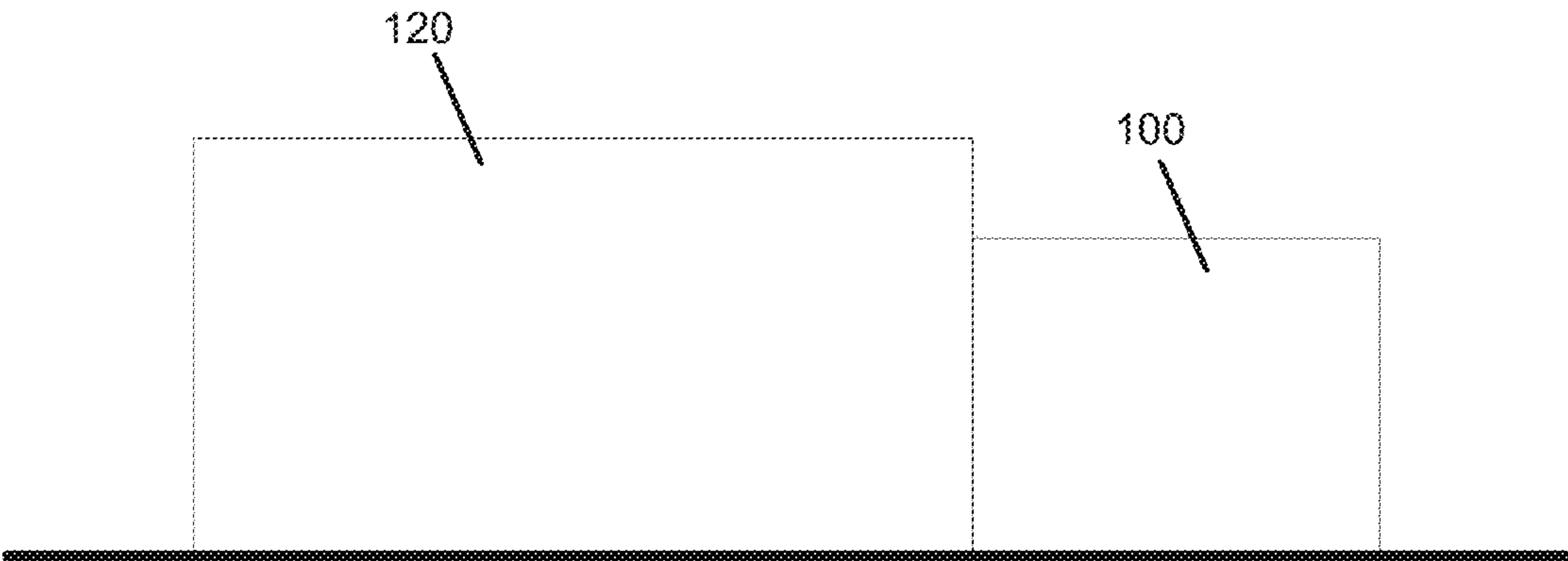


FIG. 7

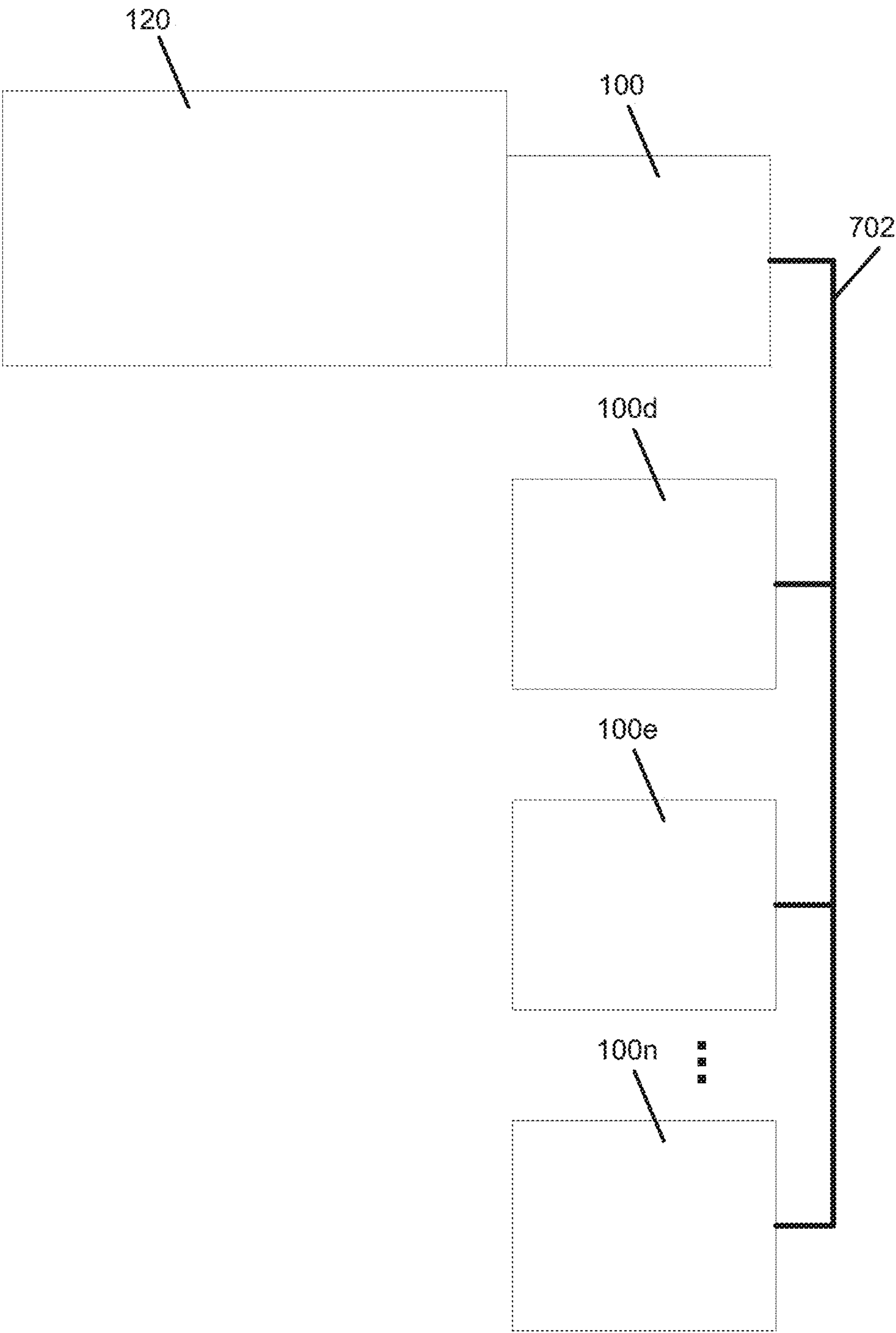
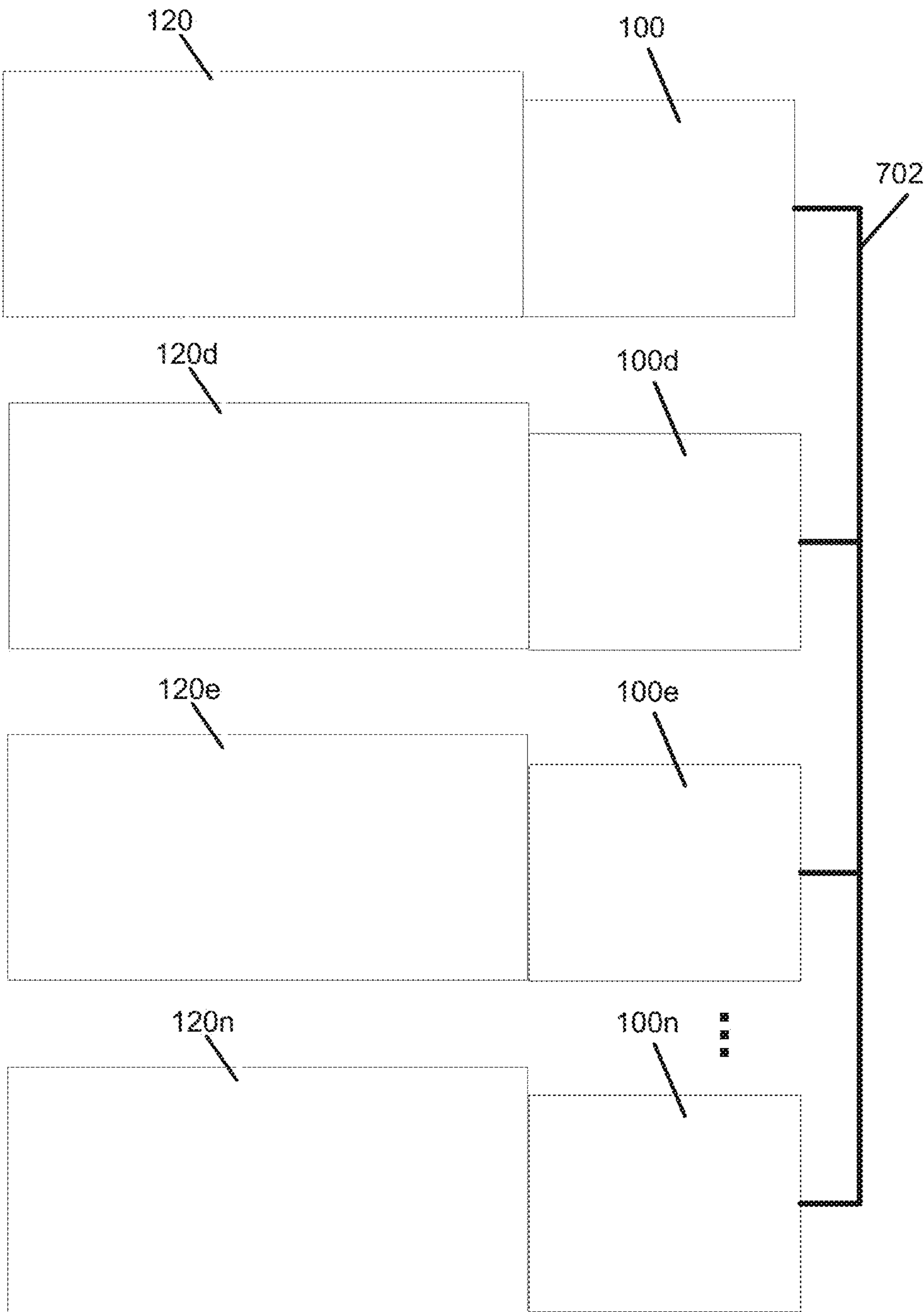


FIG. 7A



HYBRID ENERGY SYSTEMS**CROSS REFERENCE TO RELATED APPLICATIONS**

[0001] The present application claims the filing benefits of U.S. provisional application, Ser. No. 63/344,117, filed May 20, 2022, which is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

[0002] The present invention is directed to battery systems, power generation, system controls, and in particular to battery storage systems in combination with generators in mobile and stationary applications.

BACKGROUND OF THE INVENTION

[0003] Generators are often driven by engines, such as diesel engines, and other engines using a variety of fuels, e.g., propane, natural gas, gasoline, biodiesel, and hydrogen. When properly loaded, a diesel engine provides an efficient driver for a generator, when lightly loaded, diesel engines are susceptible to wet stacking (i.e., when unburnt diesel fuel passes into the diesel exhaust system and produces an oily residue). Wet stacking happens when a diesel engine is running at a low percentage or proportion of its capacity. For example, a diesel engine coupled to a generator is susceptible to wet stacking when the generator it is driving has no load or only a minimal load coupled to it; in addition the generator operates less efficiently (more fuel is consumed per kwh produced versus an optimally loaded generator). When the generator is operating with no load or only a minimal load, the diesel engine is operating in an inefficient manner, resulting in the risk of wet stacking because the diesel engine is not at a proper operating temperature (allowing unburnt fuel to escape into the diesel exhaust system). Diesel engines are most efficient when they are running at a sufficient percentage or proportion of their full capacity. When a diesel engine is running under a sufficient load, the diesel engine can run at an optimum operating temperature. Generators used to power electrical equipment typically see variable loads through their normal usage intervals. To aid in the prevention of wet stacking of diesel engines coupled to generators, dummy loads or load banks can be applied to their generators. The load banks provide a load on the generator that is sufficient to prevent wet stacking of the generator's diesel engine. However, when a load bank or an actual load is applied to increase the load on the generator, the diesel generator, while cleaner burning, will consume significantly more fuel to meet the increased generator load.

SUMMARY OF THE INVENTION

[0004] Embodiments of the present invention provide a system for managing an operational environment of an engine-driven generator. Example engines include a variety of fuel sources, for example, diesel, propane, natural gas, gasoline, biodiesel, and hydrogen. A hybrid energy system improves the efficiency of an engine-powered generator by running the generator at its optimal load and reducing the generator's run time by storing unused power, or power the generator is capable of producing that is more than the load requires, in a battery. Use of the hybrid energy system completely eliminates light loading and wetstacking issues

on the engine/generator, as well as allows the engine/generator to power down while the batteries in the hybrid energy system supplies power to the load. When the battery of the hybrid energy system is sufficiently charged, the engine/generator can be shut down to reduce engine/generator run time and the battery used to run the load until it reaches a low charge threshold (in percentage or battery voltage). The engine/generator would be turned back on to power the load when the battery charge reaches this low charge threshold. With the engine/generator powering the load, the battery can be recharged by the generator's available power output that exceeds the load. The engine/generator can also be turned on when the power load requirements are above a load threshold with respect to the generator and/or the battery. Thus, the hybrid energy system minimizes engine/generator run hours, improves fuel consumption, and reduces emissions compared to a conventional engine/generator set up to power a load.

[0005] The hybrid energy system includes a battery bank that includes a plurality of batteries that are arranged in an electrical circuit to supply a desired voltage output. The hybrid energy system may include a DC/DC converter that adjusts (raises or lowers) the voltage output of the battery bank to a desired DC voltage that is applied to a DC bus. The hybrid energy system may include an AC/DC converter that receives and converts an AC voltage from the generator to a DC voltage that is applied to the DC bus. The hybrid energy system includes a DC/AC converter that converts the DC voltage from the DC bus to an AC voltage that is applied to an AC outlet panel, which may include filters and/or transformers. The hybrid energy system may also include a DC bus interface that is coupled to the DC bus for supplying DC power to an external load and/or for supplying DC power to the DC bus from an external DC power source. The hybrid energy system powers a load while the engine/generator is powered down.

[0006] In another embodiment of the present invention, a hybrid energy system is configured to provide a load to a generator while the generator charges the hybrid energy system, and with the hybrid energy system further configured to power a load such that the generator can be powered down. The hybrid energy system includes a battery bank that includes a plurality of batteries arranged into a configuration to output a first DC voltage at a desired voltage level. The hybrid energy system may include a DC/DC converter, an AC/DC converter, and a DC/AC converter. The DC/DC converter converts the first DC voltage from the battery bank to a second DC voltage that is applied to a DC bus. The AC/DC converter converts a first AC voltage from the generator to a third DC voltage that is applied to the DC bus. The second DC voltage and the third DC voltage have about the same target voltage levels and are applied to a common DC bus. The DC/AC converter converts the DC voltage from the DC bus to a second AC voltage. The second AC voltage is applied to an AC outlet interface that provides for a variety of AC voltage levels using various step up or step down transformers to obtain the desired voltage for the AC outlet interface. The DC/DC converter supplies a fourth DC voltage to the battery bank to recharge the battery bank.

[0007] In a further embodiment of the present invention, an exemplary hybrid energy system is configured to carry a power load for a generator configured to output a first AC signal. The hybrid energy system includes a battery bank, a DC/DC converter, an AC/DC converter, and a DC/AC

converter. The battery bank includes a plurality of batteries and outputs a first DC signal. The DC/DC converter, operating in a first mode, receives and converts the first DC signal into a second DC signal, which is output to a DC bus. The AC/DC converter receives and converts the first AC signal into a third DC signal. The second DC signal and the third DC signal have a same voltage level. The DC/AC converter receives and converts the second DC signal from the DC bus into a second AC signal, which is output to an AC outlet interface.

[0008] In yet another embodiment of the present invention, an exemplary method for controlling a hybrid energy system that is carrying a power load of a generator includes outputting a first DC signal from a battery bank. The method includes receiving and converting with a DC/DC converter, the first DC signal into a second DC signal and outputting the second DC signal to a DC bus. A first AC signal, output by the generator, is received and converted by an AC/DC converter into a third DC signal. The second DC signal and the third DC signal have a same voltage level. The method also includes receiving and converting, with a DC/AC converter, the second DC signal from the DC bus into a second AC signal. The second AC signal is output to an AC outlet interface.

[0009] In an aspect of the present invention, the hybrid energy system is positioned upon a trailer or truck bed that also holds the engine/generator. In another aspect of the present invention, the engine/generator and the hybrid energy system are combined in a unified body (i.e., an “all-in-one” body rather than two separate bodies that are coupled) that may be positioned upon a trailer or truck bed or upon some other surface.

[0010] In a further aspect of the present invention, an external battery system is coupled to the DC bus via a DC bus interface. The external battery system is configured to provide DC power to the DC bus and/or to receive DC power from the DC bus via the DC bus interface.

[0011] In a further aspect of the present invention, an external generator is coupled to the DC bus via a DC bus interface and/or an AC bus via an AC bus interface. The external generator is configured to provide AC power to the AC bus via the AC bus interface and/or provide DC power to the DC bus via the DC bus interface.

[0012] In another aspect of the present invention, an exemplary hybrid energy system is configured to allow a generator to operate at an optimal or full load (approximately up to 100%) while charging batteries and powering existing loads while maximizing fuel usage for a majority of the time the generator is running (unless at warm-up and cool-down modes). This applies to both a fixed speed generator at a certain engine RPM or a variable speed generator where the engine may rev higher (which provides more output power to the common DC bus) or lower (which provides less power but may improve fuel consumption). This allows more power to be generated by the generator to offset a load that is greater than what the battery is able to output from its DC/DC converter. Higher loads on the AC outlet interface are sensed by the DC/AC converter, which causes the controller to run a variable speed generator at a higher rpm as needed to offset the demand.

[0013] In yet another aspect of the present invention, the DC/AC converter is configured to adjust the AC voltage output at the AC outlet interface up or down to produce an adjusted AC voltage with an adjusted voltage level.

[0014] In a further aspect of the present invention, the system is configured to simultaneously provide a plurality of AC signals at a plurality of voltage levels.

[0015] In another aspect of the present invention, the battery bank and an external battery bank are controlled independently, such that the battery bank and the external battery bank are charged and/or discharged at varying rates to maximize cycle life, improve overall system efficiency, and operation.

[0016] In yet another aspect of the present invention, the hybrid energy system is configured to work with any fuel-type generator, such as propane, natural gas, gasoline, ethanol, biodiesel, or hydrogen.

[0017] In a further aspect of the present invention, the hybrid energy system is configured to interface with, and to function with, a DC generator by replacing the AC/DC converter with a DC/DC converter. The DC/DC converter in this aspect would not be required if the DC generator can output a DC voltage that matches the DC bus. In this configuration, there are only two converters (i.e., a DC/DC converter for the battery and a DC/AC converter to the AC outlet interface). The AC/DC converter has been replaced with the DC generator.

[0018] In another aspect of the present invention, the controller is configured to control the recharging of the battery bank such that the generator, when running, operates at an optimal and/or full load. The controller is configured to control the discharge of the battery bank, such that an operational run time of the generator is minimized.

[0019] In yet another aspect of the present invention, the controller is configured to output a startup signal to the generator to start outputting the first AC signal when the controller selects the second mode for the DC/DC converter. The controller is configured to control the operation of the generator such that the generator warms up and cools down before shutdown. The controller controls the operation of the generator such that the generator ramps up its power output and ramps down its power output.

[0020] In a further aspect of the present invention, the hybrid energy system is paralleled with a second generator. The two generators are configured to utilize a common output bus such that the second generator may be utilized as a spinning reserve.

[0021] The exemplary system controls and the DC bus configuration allow for a smooth transfer of power to and from the generator and the battery bank. A loss of one of the input power sources (either the generator or the battery bank) will not result in an immediate loss of power.

[0022] These and other objects, advantages, purposes, and features of the present invention will become apparent upon review of the following specification in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 is a block diagram illustrating an exemplary hybrid energy system and generator coupling via converters to a DC bus in accordance with the present invention;

[0024] FIG. 2 is a perspective view of an exemplary hybrid energy system arranged alongside a generator and illustrating a DC bus access panel in accordance with the present invention;

[0025] FIG. 3 is an opposite side perspective view of the hybrid energy system and generator of FIG. 2 and illustrating an AC outlet panel;

[0026] FIG. 4 is a perspective view of the hybrid energy system and generator of FIG. 2 and illustrating an accessory outlet post in accordance with the present invention;

[0027] FIG. 5 is a perspective view of the hybrid energy system and generator of FIG. 2 and illustrating an electric vehicle cabled to the accessory outlet post for recharging in accordance with the present invention;

[0028] FIG. 6A is a perspective view of the hybrid energy system and generator of FIG. 2 and illustrating a second hybrid energy system cabled to the DC bus access panel in accordance with the present invention;

[0029] FIG. 6B is a perspective view of the hybrid energy system and generator of FIG. 2 and illustrating a second generator (or multiples of) cabled to the DC bus access panel, allowing for generator redundancy or combined power by connecting the secondary generator to its own hybrid energy system in accordance with the present invention;

[0030] FIG. 6C is a view of an exemplary solar panel array or assembly connected via a DC/DC converter for coupling to an exemplary hybrid energy system via the DC bus access panel in accordance with the present invention;

[0031] FIG. 6D is a view of the hybrid energy system of FIG. 6C positioned upon a trailer and integrated with a solar panel array in accordance with the present invention;

[0032] FIG. 6E is a perspective view of a variety of hybrid energy system and generator configurations in accordance with the present invention;

[0033] FIG. 6F is a block diagram of a hybrid energy system and generator arranged together within a unitary housing for stationary operation in accordance with the present invention;

[0034] FIG. 6G is a block diagram of the hybrid energy system and generator of FIG. 6E arranged on a trailer in accordance with the present invention;

[0035] FIG. 6H is a block diagram of an exemplary hybrid energy system and generator arranged for stationary operation in accordance with the present invention;

[0036] FIG. 7 is a block diagram of an exemplary hybrid energy system and generator arranged in parallel with a plurality of hybrid energy systems in accordance with the present invention; and

[0037] FIG. 7A is a block diagram of an alternative hybrid energy system and generator arranged in parallel with a plurality of hybrid energy systems and associated generators in accordance with the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0038] Referring to the drawings and the illustrative embodiments depicted therein, a hybrid energy system provides for the elimination or reduction of loading issues on engine driven generators. Such engines may be powered by a variety of fuels, for example, diesel, propane, natural gas, gasoline, biodiesel, and hydrogen. The engine/generators include, for example, exemplary diesel engine driven generators, such as, for example, EPA Tier 4F certified or other similarly compliant diesel engine driven mobile generators (as well as Stage 5 or higher). The hybrid energy system also provides for the powering down of the diesel engine driven generators while the hybrid energy system provides power to a load. Exemplary hybrid energy systems may be either unified or arranged alongside portable prime power skid generators or trailer-mounted diesel generators and provide

selective AC and/or DC power output to power a load while the generator is powered down. Alternatively, the hybrid energy system is designed for a stationary application as well as a mobile application. Alternatively, the hybrid energy system provides an additional power load (the unused power is used to recharge the battery bank of the hybrid energy system) to an associated generator when a real load is below a threshold load value. Most generators are more efficient at a full load or close to it where a generator can provide more kilowatts per volume of fuel consumed compared to being light loaded where efficiency is decreased. Light loading certain generators, such as diesel generators, introduces engine and exhaust problems with unburned fuel that require additional generator maintenance and repair. Powering a load with a generator running at its optimal or full load and utilizing any or all remaining power from the generator to charge batteries that can be used to power the load while the generator is off greatly improves the generator's efficiency and reduces overall fuel consumption, emissions, and run hours. Exemplary embodiments discussed herein pertain to temporary or portable power generation and energy storage that can be transported from one location to another or may be a permanent installation.

[0039] FIG. 1 illustrates a hybrid energy system 100 that includes an exemplary AC generator 120 and a battery bank 102. An exemplary AC/DC converter (active front end "AFE") 108 is electrically coupled to the AC generator 120, while a bidirectional DC/DC converter 104 is electrically coupled to the battery bank. A DC bus is configured to electrically couple the AC/DC converter 108 and the bidirectional DC/DC converter 104a. A DC/AC converter 110 and a DC bus interface 114 are electrically coupled to the DC bus 106. As illustrated in FIG. 1, an AC load 101a is coupled to the AC generator 120, a second AC load 101b is coupled to AC outlets 116a-n of an AC outlet interface 112, which is electrically coupled to the DC/AC converter 110. FIG. 1 also illustrates external DC loads 115a-n and/or external DC power supplies 117a-n coupled to the DC bus 106 via the DC bus interface 114 to connect to multiple energy trailers (e.g., an external battery bank), or other DC devices.

[0040] As illustrated in FIG. 1, the hybrid energy system 100 is configured to receive power from the AC generator 120, which is configured to output an AC voltage (e.g., three-phase 480 VAC). The AC/DC converter 108, which is electrically coupled to the AC generator 120, is configured to convert the generator's AC voltage output to a selected DC voltage that is applied to the DC bus 106. In an aspect of the present embodiment, the AC/DC converter 108 is an exemplary active front end (AFE) inverter and includes power magnetics (filters) and associated hardware and software (see FIGS. 3-5) to convert an AC signal to a DC signal. An exemplary AC/DC converter 108 is manufactured using insulated-gate bipolar transistor (IGBT) technology. In a further aspect of the present embodiment, the AC/DC converter 108 (implemented as an active front end inverter) is configured to eliminate harmonics and to provide a power factor for the AC generator 120 of unity (1.0), or near unity, in an ideal case where 100% of the current output from the AC generator 120 contributes to real power in the load. In an aspect of the present embodiment, the AC/DC converter 108 is configured to provide a power factor of 0.95 to 1.0. When the exemplary AC/DC converter 108 is achieving a power factor of unity, or near unity, the current output of the

AC generator **120** has a real power output that matches its apparent power output. Thus, the AC generator **120** meets the required load power output with a lower total current draw. This allows for a simpler and less expensive diesel generator build because of the lower total current draw requirement. The exemplary AC/DC converter **108** is an improvement over conventional engine-powered generators and AC/DC inverter assemblies, which are required to deal with undesirable harmonics and a lower power factor that are put back on the engine-powered generator.

[0041] The DC voltage output by the AC/DC converter **108** is approximately the same voltage level as the DC voltage output by the bidirectional DC/DC converter **104** (e.g., approximately 700 VDC). For example, in one exemplary embodiment, the AC/DC converter **108** and the bidirectional DC/DC converter **104** are both configured to output a DC voltage of approximately 700 VDC. That is, the DC voltages output by the AC/converter **108** and the DC/DC converter **104** are within $\pm 10\%$ of the desired voltage level. In other embodiments, the converters could be configured for different voltage outputs.

[0042] The battery bank **102**, which includes a plurality of batteries **110a-n**, is configured to output a selected DC voltage (see FIG. 2B). In one exemplary embodiment, the battery bank **102** is configured to output a DC voltage (e.g., 250-400 VDC). With the addition of more batteries **110**, a higher output DC voltage (up to the maximum DC bus voltage) is possible by arranging batteries in a series configuration. There is no system predefined limit to the kWh as additional batteries can be connected in parallel without impacting the DC battery voltage—as long as the battery management system (BMS) is configured and capable of handling the parallel connections.

[0043] The battery bank **102** is coupled to the bidirectional DC/DC converter **104** that is configured to convert the DC voltage output by the battery bank **102** to a selected DC voltage that is applied to the DC bus **106**. In one exemplary embodiment, the 250-400 VDC output from the battery bank **102** is increased by the bidirectional DC/DC converter **104** to approximately 700 VDC. In an aspect of the present embodiment, the bidirectional DC/DC converter **104** is an exemplary bidirectional DC-to-DC converter that includes power magnetics (filters) and associated hardware and software (see FIGS. 3-5) to convert the DC signal into a different DC signal with a different voltage level. An exemplary bidirectional DC/DC converter **104** is manufactured using insulated-gate bipolar transistor (IGBT) technology. The bidirectional DC/DC converter **104** is configured as a dual mode or two-way converter for either discharging the battery bank **102** (while powering a load) or recharging the battery bank **102**. In an aspect of the present embodiment, the bidirectional DC/DC converter **104** is configured to automatically reverse from outputting a DC voltage from the battery bank **102** to begin recharging the battery bank **102** when the battery bank's output voltage/current drops below a threshold set point. The AC generator **120** is signaled to start either from a configurable state of charge reading of the battery bank **102** from an internal BMS, or a pre-defined load limit. These are exemplary programmable limits.

[0044] When operating in the second mode (or battery recharge mode), the bidirectional DC/DC converter **104** receives the approximately 700 VDC voltage output from the DC bus **106** and converts it (down) to the programmable charging voltage of the battery bank **102** (which will vary

based on the number of batteries in the battery bank **102**). This DC voltage is received by the battery bank **102** during the recharge mode. As discussed herein, when operating in the recharge mode, the bidirectional DC/DC converter **104** allows the AC generator **120** to operate at an optimal efficiency (at or near 100% power output). By recharging the battery bank **102** at a full or nearly full current output, the AC generator **120** is allowed to run most efficiently and to maximize its fuel efficiency.

[0045] The DC/AC converter **110** is coupled to the DC bus **106** and is configured to convert the DC voltage of the DC bus **106** (e.g., approximately 700 VDC) to an AC voltage that is applied to the AC outlet interface **112**. In an aspect of the present embodiment, an exemplary DC/AC converter **110** is a DC to AC output inverter and is configured to simultaneously output multiple AC voltages, for example: single phase 120V and 240V, and three-phase 120V/208V and 277V/480V. Such voltage, phase, and frequency selections includes voltages and frequencies for use in Europe and Canada, as well as other applications that require different voltages and frequencies that the DC/AC converter can output. The DC/AC converter **110** includes transformers and associated hardware and software (see FIGS. 3-5) to convert the DC signal into the multiple AC signals. An exemplary DC/AC converter **110** is manufactured using insulated-gate bipolar transistor (IGBT) technology. The DC/AC converter **110** outputs the variety of AC voltage levels to respective AC outlets **116a-n** of the AC outlet interface **112**. Each of the exemplary AC voltage levels is supplied to one or more corresponding AC outlets **116a-n**. For example, the AC outlet interface **112** can include two (or more) AC outlets **116a,b** outputting 120 VAC, while only including single AC outlets **116c** and **116d**, each outputting 277 VAC or 480 VAC, respectively. Each of these AC voltage outputs from the AC outlets **116a-n** are received by the AC load **101b**. As illustrated in FIG. 1, in addition to the AC load **101b** coupled to the AC outlets **116** of the AC outlet interface **112**, the separate AC load **101a** is coupled to the AC generator **120**. The AC load **101a** and the AC/DC converter **108** are arranged in a parallel circuit with the AC generator **120**. While an exemplary load bank may be used to provide an additional loading on the AC generator **120**, the AC loads **101a,b** are considered “real” loads demanding AC power. Load banks are described in detail in U.S. patent application Ser. No. 17/668,579, which is hereby incorporated by reference herein in its entirety.

[0046] As also illustrated in FIG. 1, the hybrid energy system **100** includes the DC bus interface **114**. The DC bus interface **114** is configured to provide external access to the DC bus **106**. As discussed herein, one or more external DC power sources **117a-n** may be coupled to the DC bus **106** via the DC bus interface **114** to supply additional DC power to the DC bus **106**. Alternatively, one or more external DC loads **115a-n** can be coupled to the DC bus **106** via the DC bus interface **114** to receive DC power from the DC bus **106**. The DC bus interface **114** may also include an exemplary bidirectional DC/DC converter **116** for converting the DC voltage on the DC bus **106** to a desired DC voltage output for the external DC load **115a-n** or for converting a DC voltage supplied by the external DC power source **117a-n** to a desired DC voltage level to be applied to the DC bus **106**.

[0047] As illustrated in FIG. 1, the bidirectional DC/DC converter **104** selectively converts the DC voltage of the DC bus **106** to a desired DC voltage for recharging the batteries

110a-n of the battery bank **102**. A controller **118**, which is communicatively coupled to the battery bank **102** and the bidirectional DC/DC converter **104**, controls the output of the battery bank **102** and controls the operation of the bidirectional DC/DC converter **104** (e.g., selective recharging of the battery bank **102**). The bidirectional DC/DC converter **104** reacts to the DC bus voltage level—when it is above a setpoint, the batteries charge, and when it is below another setpoint, the batteries discharge. The controller **118** can dynamically change that setpoint to allow the independent load and source control of multiple energy sources (generators) or battery systems. The rate of charge of the batteries is dynamically controlled by the controller **118** and the DC/DC converter **104**.

[0048] The controller **118** also controls the operation of the AC generator **120**. For example, the controller **118** outputs a startup signal or shutdown signal to the AC generator **120** for starting and stopping the AC generator **120**, respectively. The controller **118** controls the operation of the AC generator **120**, such that the AC generator **120** can be warmed up at startup and cooled down before shutdown. Such control at startup and shut down by the controller **118** may include ramping up the power output of the AC generator **120** during startup and ramping down the power output of the AC generator **120** during shutdown. The controller **118** is also configured to control the recharging of the battery bank **102** such that the AC generator **120**, when running, operates at an optimal (i.e., for fuel efficiency and operational conditions) and/or full load. The controller **118** may also be used to control the discharge of the battery bank **102** such that an operational run-time of the AC generator **120** is minimized. That is, by powering a load applied to the system with power from the battery bank **102** (when sufficiently charged as discussed herein), the AC generator **120** may be shutdown to save fuel and runtime.

[0049] Thus, the AC generator **120** and the hybrid energy system **100** (as controlled by the controller **118**) may operate in a variety of modes. For example, in a first mode of operation, the AC generator **120** outputs AC voltage to power the AC load **101a**, while the controller **118** selectively shuts down the AC/DC converter, such that the AC generator **120** does not supply power to the DC bus **106**. In the case of a DC generator, there is no AC/DC converter, so the DC generator shuts down and does not supply power to the DC bus **106**. The AC generator **120** may be in the first mode of operation when the AC generator **120** is supplying power to an AC load **101a** that is above a power threshold with respect to a maximum power output of the AC generator **120**. In the first mode of operation, the AC generator **120** supplies power to the AC load **101a** while not providing power to either the AC outlet interface **112** or providing “recharging” power to the battery bank **102**. In the first mode of operation, when the battery bank **102** has a sufficient charge, DC voltage may be output by the battery bank **102** to power external DC loads **115a-n** (via the DC bus interface **114**) and/or converted to AC signals via the DC/AC converter **110** to power an AC load **101b** via the AC outlet interface **112**. The exemplary AC generator **120** may be configured as a fixed speed generator (at a selected engine RPM) or a variable speed generator where the engine may increase its RPM rate (which provides more output power to the DC bus **106**). Such arrangement allows more power to be generated by the generator **120** to offset a load that is greater than what the battery **102** is able to output from its

DC/DC converter **104**. Higher loads on the AC outlet interface **112** are sensed by the DC/AC converter **110**, which causes the controller **118** to run a variable speed generator at a higher RPM as needed to offset the power demand.

[0050] In a second mode of operation, the AC generator **120** outputs AC voltage to the AC/DC converter **108** to supply power to the AC load **101a** and/or the AC load **101b** (via the DC bus **106** and the DC/AC converter **110**). In the second mode of operation, the controller **118** has also deactivated the bidirectional DC/DC converter **104**, such that AC generator **120** does not provide “recharging” power to the battery bank **102** because there is no or an insufficient quantity of “unused” power. When “deactivated,” the bidirectional DC/DC converter **104** is not “shut off,” it just no longer charges or discharges the battery bank **102**.

[0051] In a third mode of operation, the AC generator **120** outputs AC voltage to the AC/DC converter **108** to supply power to the AC load **101a** and/or the AC load **101b** (via the DC bus **106** and the DC/AC converter **110**). In the third mode of operation, the controller engages the bidirectional DC/DC converter **104** in a recharge mode such that the bidirectional DC/DC converter **104** provides “recharging” power to recharge the battery bank **102**. The selective recharging of the battery bank **102** is controlled by the controller **118**. Whether the battery bank **102** is recharged is dependent upon several factors that are monitored by the controller **118**. For example, the controller **118** can allow the battery bank **102** to be recharged when a charge level of the battery bank **102** is below a first charge threshold, while stopping the recharging of the battery bank **102** when the charge level of the battery bank **102** is above a second charge threshold. The controller **118** can also selectively recharge the battery bank **102** depending on the AC loads **101a,b** coupled to the AC generator **120** and/or the AC outlet interface **112**, and depending on the external DC loads **115a-n** coupled to the DC bus interface **114**.

[0052] As discussed herein, the AC generator **120** can be used to selectively recharge the battery bank **102** such that the AC generator **120** is allowed to work under a loaded condition (even when there is no load or only a minimal load coupled to the AC generator **120**) that is sufficient to prevent wet stacking when variable real loads (e.g., AC loads **101a**, **101b**, and DC loads **115a-n**) are too low to prevent wet stacking. Wet stacking can occur when the diesel engine is run below an optimal operating temperature.

[0053] FIG. 2 illustrates an exemplary hybrid energy system **100** arranged on a trailer **202** alongside an exemplary AC generator **120**. As illustrated in FIG. 2, the hybrid energy system **100** includes a DC bus interface **114** with a plurality of exemplary power cords **602** running to exemplary DC loads **115** or DC power supplies **117** (see FIGS. 5 and 6A-6C).

[0054] FIG. 3 illustrates an opposite side of the hybrid energy system **100** and AC generator **120** of FIG. 2. As illustrated in FIG. 3, the hybrid energy system **100** includes an AC outlet interface **112** with a plurality of AC outlets **116**. FIG. 3 also illustrates an exemplary programmable control panel **702** for interfacing with the hybrid energy system **100**. For example, the control panel **702** allows user interaction with the controller **118** of the hybrid energy system **100**. The control panel **702** includes a programmable graphical user interface (e.g., a touch-responsive panel) for inputting user feedback. The control panel **702** displays basic generator information and operational status, which can be monitored

and reported. The control panel **702** also displays basic battery and converter/inverter information from the hybrid energy system **100**.

[0055] FIG. 4 illustrates the hybrid energy system **100** and AC generator **120** of FIG. 2 with the addition of an exemplary DC outlet box **802**. As illustrated in FIG. 4, the DC outlet box **802** is electrically coupled to the DC bus interface **114** via an exemplary cable **804**.

[0056] FIG. 5 illustrates the exemplary hybrid energy system **100** and AC generator **120** of FIG. 2, which includes the DC outlet box **802** of FIG. 4. FIG. 5 illustrates an exemplary DC load **115a** cabled to the DC outlet box **802** for recharging an onboard battery. While FIG. 5 illustrates an exemplary battery powered forklift as the DC load, other industrial battery powered equipment may be electrically coupled via the DC outlet box **802** for recharging. In an aspect of the present embodiment, one or more DC loads **115** may be coupled to the DC outlet box **802**. The DC outlet box **802** is configured to provide DC power for powering and/or recharging. As illustrated in FIG. 6D, the exemplary hybrid energy systems **100** and AC generators **120** may be sized for an expected load need.

[0057] In FIG. 6A, an exemplary external power supply **117a** is coupled to the DC bus interface **114**. In one exemplary embodiment, the power supply **117a** illustrated in FIG. 6A is an exemplary external battery bank **117a**. The external battery bank **117a** may be similar to the battery bank **102** or a different configuration or using different energy storage technology or battery chemistry. Referring to FIG. 7, the external power supply **117a** (of FIG. 6A) is replaced with a second hybrid energy system **100d** or a plurality of hybrid energy systems **100d-100n**. The controller **118** is configured to control the plurality of hybrid energy systems (**100**, **100d-100n**). The output of each of the plurality of hybrid energy systems **100**, **100d-100d** are tied together using a common bus **702** communicatively coupled to the AC output interfaces of at least two or more of the hybrid energy systems **100**, **100d-100n**, such that the plurality of hybrid energy systems **100**, **100d-100n** can output in parallel both three-phase and single-phase AC power. In FIG. 7A, an alternative arrangement of FIG. 7, each of the hybrid energy systems **100d-100n** is paired with an associated generator **120d-100n**. The parallel arrangement illustrated in FIG. 7A allows for additional flexibility with multiple generators **120**, **120d-120n** selectively used to power a load or recharge a battery bank **102** in an own hybrid energy system **100** or another parallel-configured hybrid energy system **100d-100n**.

[0058] In FIG. 6B, an exemplary external power supply **117b** is coupled to the DC bus interface **114**, via an external power distribution box **1002** which is configured to provide a redundant interface for coupling a load to the generator **120** and the hybrid energy system **100**. In one exemplary embodiment, the power supply **117b** illustrated in FIG. 6B is an external AC generator **117b**. The external AC generator **117b** may be similar to the AC generator **120** or a different configuration or using different AC power generation technology. In one aspect of the present embodiment, the external power source outputs a DC voltage via an AC/DC converter in the external AC generator **117b**. The hybrid energy system **100** may also be paralleled with the external AC generator **117b**. That is, the AC generator **120** and the

external AC generator **117b** may utilize a common output bus such that the external AC generator **117b** may be utilized as a spinning reserve.

[0059] In a further aspect of the present invention, the hybrid energy system is paralleled with a second generator. The two generators are configured to utilize a common output bus such that the second generator may be utilized as a spinning reserve.

[0060] In FIG. 6C, an exemplary external power supply **117c** is coupled to the DC bus interface **114**. In one exemplary embodiment illustrated in FIG. 6C, the power supply **117c** is an exemplary solar panel array **117c**. Other alternative renewable energy sources are also considered, such as, wind-driven turbines, fuel cell technology, and other energy sources. The alternative power sources of FIGS. 6A-6H provide redundancy of energy storage or generation. The external power distribution box **1002** and the DC outlet box **802** provide redundant interfaces for coupling a load to the generator **120** and the hybrid energy system **100**. In FIG. 6D, an exemplary on-board solar panel **117c** and solar inverter **150** are plugged into an AC outlet interface **112** providing power to a load and charging batteries **102** of the hybrid energy system **100** (through either of the DC/DC converters **104**, **116** of the hybrid energy system **100**) when loads are below the output of the solar panels **117c**.

[0061] As illustrated in FIGS. 6A-6H, a plurality of hybrid energy systems **100** may be coupled together to run in parallel. Each output of the plurality of hybrid energy systems **100** are tied together using a common bus communicatively coupled to the AC output interfaces of two or more hybrid energy systems **100** of the plurality of hybrid energy systems, such that the plurality of hybrid energy systems output in parallel both three-phase and single-phase AC power. As also illustrated in FIGS. 6A-6H, the hybrid energy systems **100** and generators **120** may be configured in various sizes and shapes. For example, the hybrid energy systems **100** and generators of FIGS. 6F-6H (as well as others illustrated in FIGS. 6A-6E) are not drawn to scale.

[0062] FIG. 6E illustrates a trio of exemplary embodiments with varying sizes of hybrid energy systems **100a**, **100b**, and **100c**, and corresponding varying sizes of AC generators **120a**, **120b**, and **120c**. As illustrated in FIG. 6D, exemplary embodiments may be sized to fit a desired operational environment and AC and/or DC power load needs. In FIG. 6F, a hybrid energy system **100** and a generator **120** are arranged within a unitary body **130**. Such an arrangement **130** may be positioned upon a trailer **202** (see FIG. 6G) or configured as free-standing or stationary (FIG. 6F). Such coupled versions (with separate enclosures for the hybrid energy system **100** and the generator **120**) may also be arranged as free-standing or stationary (see FIG. 6H). The unitary body arrangement **130** may be substituted for any of the embodiments with generators **120** and hybrid energy systems **100** arranged in separate housings. For example, the generators **120d-120n** and hybrid energy systems **100d-100n** of FIG. 7A, which are illustrated as being arranged within separate housings, could be replaced with the generator/hybrid energy system combinations arranged in unitary bodies **130**. In another exemplary embodiment, a combination of unitary bodies **130** (each housing a generator/hybrid energy system **120** arrangement) could be used together with one or more generators **120** and hybrid energy systems **100** in separate housings.

[0063] While the control panel 702 (see FIG. 3) displays hybrid energy system voltage outputs and operating parameters, in an aspect of the embodiments discussed herein, output voltage and operating parameters can be selected from the control panel 702. The selected voltage can also be fine-tuned and adjusted up or down by up to 10% by the user via the control panel 702. For example, an exemplary user may require an output voltage of 210 VAC instead of 208 VAC. Such adjustment may be necessary when the electrical device drawing power from the hybrid energy system 100 is a distance from the hybrid energy system 100 and experiencing a resulting voltage drop.

[0064] The control panel 702 also allows for the control of the DC bus interface 114 and any peripheral attached thereto (e.g., electrical devices coupling directly to the DC bus interface 114 or via the DC outlet box 802).

[0065] Access to the DC bus 106 (via the DC bus interface 114) allows for the connection of, for example, level 3 DC electrical vehicles or equipment, or for fast charging of electrical equipment. Such DC voltage provision can be used for mobile or temporary energy delivery applications.

[0066] Exemplary hybrid energy systems 100 and diesel generators 120 may be utility mounted to the bed of a pickup truck or towed on a trailer to be used as a mobile vehicle charger for EV vehicles or construction equipment and providing, for example, level 3 DC charging up to an exemplary 250 kilowatts or higher. Exemplary electrical equipment includes construction elevators, rechargeable construction equipment, welding equipment, pumps, job trailers, lifts and cranes, location events and/or films, and the use of remote communication equipment, such as cell towers and microwave equipment where refueling is more difficult.

[0067] In an alternative embodiment, the hybrid energy system 100 is configured to receive power from a DC generator 122 (replacing the AC generator 120 with a DC generator 122) (see FIG. 1). Subsequently, the AC/DC converter 108 would be replaced with a DC/DC converter (similar to the DC/DC converter 104). This additional DC/DC converter would be configured to receive the DC voltage output of the DC generator and convert it to the required voltage level for the DC bus 106. Note that the AC/DC converter 108 can be eliminated (when replacing the AC generator 120 with a DC generator 122) if the DC generator 122 contains an alternator that matches the voltage level of the DC bus 106.

[0068] Thus, the exemplary embodiments discussed herein improve the efficiency of a diesel generator by running it at its optimal load and reducing its run time by storing unused generated power to a battery. When the battery bank of the hybrid energy system is sufficiently charged, the generator can be shut down to reduce run time and the battery bank of the hybrid energy system may run the load entirely until it reaches a charge threshold limit. The generator would be turned back on when the battery bank's charge level reaches the threshold limit and provides power to the load while using any unused power to recharge the battery bank. The generator could also be turned on regardless of the battery bank's charge level when the load approaches either the generator's load capacity or the battery bank's load capacity. The hybrid energy system minimizes generator run hours, improve fuel consumption, and reduces emissions compared to a conventional generator setup to power a load. If using a variable speed generator, rpm may

increase as needed to provide more power if the output load is greater than the output of the DC/DC converter.

[0069] Changes and modifications in the specifically described embodiments can be carried out without departing from the principles of the present invention which is intended to be limited only by the scope of the appended claims, as interpreted according to the principles of patent law including the doctrine of equivalents.

1. A hybrid energy system configured to carry a power load for a generator configured to output a first AC signal, the hybrid energy system comprising:

- a battery bank comprising a plurality of batteries and configured to output a first DC signal;
- a DC/DC converter, when operating in a first mode, is configured to receive and convert the first DC signal into a second DC signal, wherein the second DC signal is output to a DC bus;
- an AC/DC converter configured to receive and convert the first AC signal from the generator into a third DC signal, wherein the second DC signal and the third DC signal are tied together on the DC bus; and
- a DC/AC converter configured to receive and convert the second DC signal from the DC bus into a second AC signal, wherein the second AC signal is output to an AC outlet interface.

2. The hybrid energy system of claim 1 further comprising a controller configured to control the DC/DC converter, wherein the controller is operable to adjust a voltage level of the second DC signal.

3. The hybrid energy system of claim 2, wherein the controller is configured to output a shutdown signal to the generator to stop outputting the first AC signal when the controller selects the first mode for the DC/DC converter.

4. The hybrid energy system of claim 2, wherein the controller is operable to select a second mode for the DC/DC converter, wherein the DC/DC converter, when operating in the second mode, is configured to receive and convert the second DC signal from the DC bus into a fourth DC signal, wherein the fourth DC signal is output to the battery bank to recharge the battery bank, wherein the first mode is a battery discharge mode and the second mode is a battery recharge mode.

5. The hybrid energy system of claim 4, wherein the controller is configured to control the recharging of the battery bank such that the generator, when running, operates at an optimal and/or full load, and wherein the controller is configured to control the discharge of the battery bank, such that an operational run time of the generator is minimized.

6. The hybrid energy system of claim 4, wherein the controller is configured to output a startup signal to the generator to start outputting the first AC signal when the controller selects the second mode for the DC/DC converter, wherein the controller is configured to control the operation of the generator such that the generator warms up and cools down before shutdown, and wherein the controller is configured to control the operation of the generator such that the generator ramps up its power output and ramps down is power output.

7. The hybrid energy system of claim 1 further comprising a DC bus interface configured to couple to the DC bus, wherein the DC bus interface is configured to electrically couple an electrical device to the DC bus such that the DC bus outputs a fifth DC voltage to the electrical device via the DC bus interface, wherein the electrical device is one or

more of electrical vehicle chargers, battery-powered construction equipment, electrical equipment, and consumer electrical devices.

8. The hybrid energy system of claim **7**, wherein the DC bus interface is configured to electrically couple an external power source to the DC bus such that the external power source outputs a sixth DC voltage to the DC bus via the DC bus interface, wherein the external power source comprises one or more of external battery banks, solar panel energy sources, wind-powered generators, and secondary (backup) generators.

9. The hybrid energy system of claim **1**, wherein the system is configured to simultaneously output a plurality of AC signals, wherein the plurality of AC signals comprises one or more of single-phase voltages of 120V and/or 240V, and three-phase voltages of 120V/208V and/or 277V/480V, and wherein the plurality of AC signals comprises voltages and frequencies for use in Europe and Canada, as well as other applications that require different voltages and frequencies that the DC/AC converter can output.

10. A method for controlling a hybrid energy system that is carrying a power load of a generator, wherein the method comprises:

- outputting, with a battery bank, a first DC signal;
- receiving and converting, with a DC/DC converter, the first DC signal into a second DC signal and outputting the second DC signal to a DC bus;
- receiving and converting, with an AC/DC converter, a first AC signal output by the generator into a third DC signal, wherein the second DC signal and the third DC signal are tied together on the DC bus; and
- receiving and converting, with a DC/AC converter, the second DC signal from the DC bus into a second AC signal, wherein the second AC signal is output to an AC outlet interface.

11. The method of claim **10** further comprising adjusting the DC/DC converter to adjust a voltage level of the second DC signal.

12. The method of claim **10** further comprising adjusting the AC/DC converter to adjust the voltage level of the third DC signal.

13. The method of claim **10** further comprising adjusting the DC/AC converter to adjust the voltage level of the second AC signal.

14. The method of claim **10** further comprising operating the DC/DC converter in one of a plurality of modes, wherein a first mode comprises the hybrid energy system outputting a shutdown signal to the generator to stop outputting the first AC signal.

15. The method of claim **14**, wherein a second mode comprises the hybrid energy system receiving and converting the second DC signal from the DC bus into a fourth DC

signal and outputting the fourth DC signal to the battery bank to recharge the battery bank, wherein the second mode further comprises the hybrid energy system outputting a startup signal to the generator to start outputting the first AC signal, wherein the first mode is a battery discharge mode, and the second mode is a battery recharge mode.

16. The method of claim **10** further comprising electrically coupling an electrical device to the DC bus via a DC bus interface that is electrically coupled to the DC bus, wherein the DC bus outputs a fifth DC voltage to the electrical device via the DC bus interface, wherein the electrical device is one or more of electrical vehicle chargers, battery-powered construction equipment, electrical equipment, and consumer electrical devices.

17. The method of claim **10** further comprising electrically coupling an external power source to the DC bus via the DC bus interface, wherein the external power source outputs a sixth DC voltage to the DC bus via the DC bus interface, wherein the external power source comprises one or more of external battery banks, solar panel energy sources, wind-powered generators, and secondary (backup) generators.

18. The method of claim **17**, wherein the external power source is used to recharge the battery bank via the DC bus.

19. The method of claim **10**, wherein the DC/AC converter simultaneously outputs a plurality of AC signals, wherein the plurality of AC signals comprises one or more of single-phase voltages of 120V and/or 240V, and three-phase voltages of 120V/208V and/or 277V/480V, and wherein the plurality of AC signals comprises voltages and frequencies for use in Europe and Canada, as well as other applications that require different voltages and frequencies that the DC/AC converter can output.

20. The method of claim **10** further comprising controlling a plurality of hybrid energy systems, wherein the output of each of the plurality of hybrid energy systems are tied together using a common bus communicatively coupled to the AC output interfaces of two or more hybrid energy systems of the plurality of hybrid energy systems such that the plurality of hybrid energy systems output in parallel both three-phase and single-phase AC, wherein the method further comprises the hybrid energy system paralleled with a second generator, and wherein the two generators utilize a common output bus such that the second generator is utilized as a spinning reserve.

21. The method of claim **20**, wherein while the generator is powered down, the hybrid energy system continues to power the load with one or more simultaneous output voltages, and wherein the generator is available for service or maintenance while the hybrid energy system powers the load.

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